

The Chemistry of Solutions

From Microscopic Interactions to Macroscopic Phenomena

In the natural world, pure substances are rare anomalies. Life and utility depend on mixtures. Whether it is the air we breathe, the alloys we build with, or the fluids that sustain life, understanding the science of solutions is essential to understanding matter itself.



Brass
(Solid Solution)



Bronze
(Cu + Sn)



Blood Plasma
Compatibility



Defining the Solution Landscape

The solvent determines the physical state.

The Solution Matrix

	Solute State	Solvent State	Common Example
Gaseous Solution	Gas	Gas	Mixture of oxygen and nitrogen
	Liquid	Gas	Chloroform mixed with nitrogen gas
	Solid	Gas	Camphor in nitrogen gas
Liquid Solution	Gas	Liquid	Aerated drinks (oxygen in water)
	Liquid	Liquid	Ethanol dissolved in water
	Solid	Liquid	Glucose dissolved in water
Solid Solution	Gas	Solid	Hydrogen in palladium
	Liquid	Solid	Amalgam of mercury with sodium
	Solid	Solid	Copper dissolved in gold

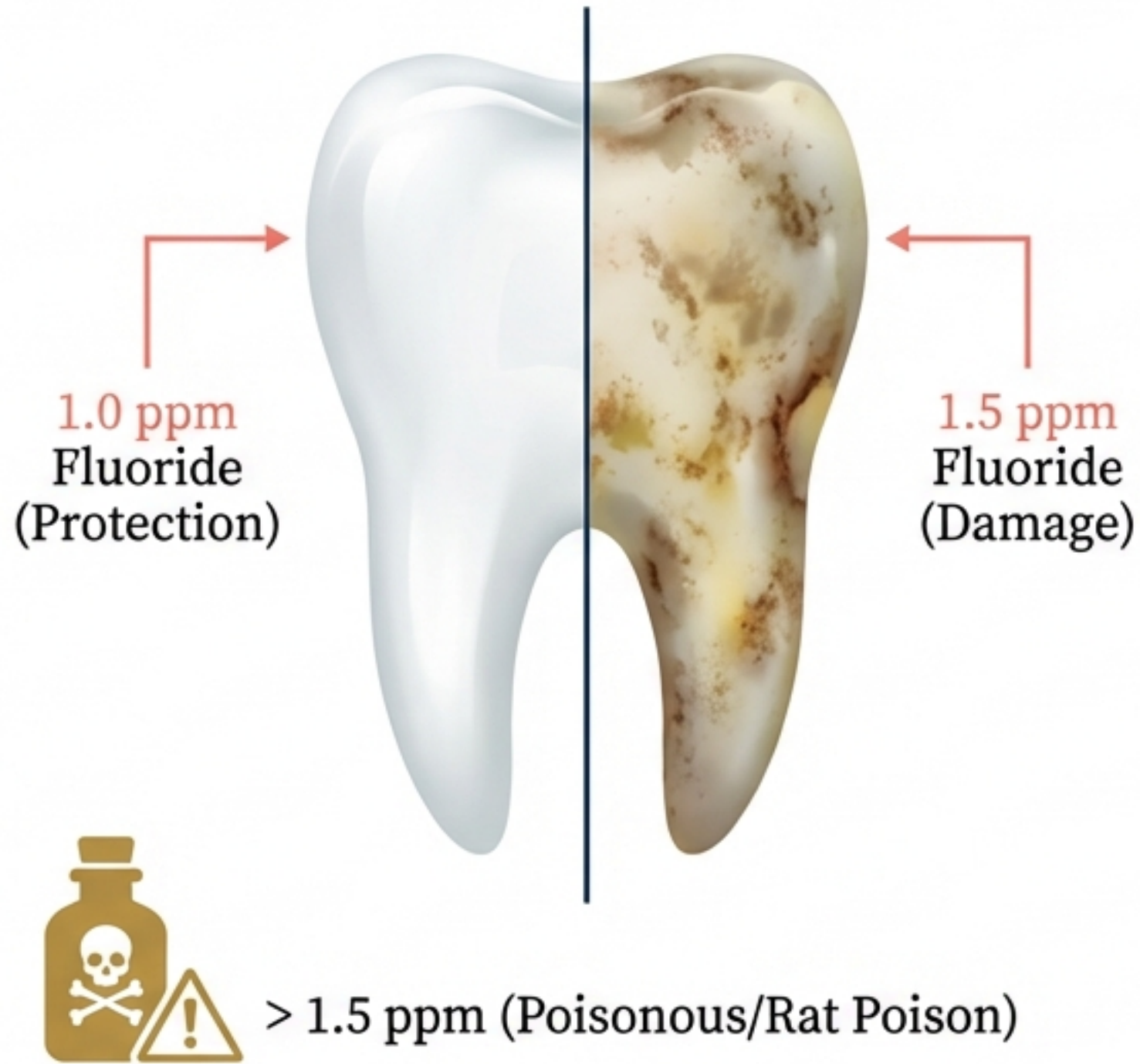
Definitions:

Solvent: The component present in the largest quantity. It determines the state.

Solute: One or more components present in the solution other than the solvent.

Quantifying Composition: Physical Units

Precision is a matter of safety.



Mass Percentage (w/w)

$$\text{Mass \%} = \frac{\text{Mass of component}}{\text{Total mass of solution}} \times 100$$

Bleach: 3.62% Sodium Hypochlorite in water.

Volume Percentage (V/V)

$$\text{Volume \%} = \frac{\text{Volume of component}}{\text{Total volume of solution}} \times 100$$

Antifreeze: 35% Ethylene glycol lowers freezing point to -17.6°C .

Parts Per Million (ppm)

$$\text{ppm} = \frac{\text{Number of parts of component}}{\text{Total parts of solution}} \times 10^6$$

Pollutants in water or atmosphere (trace quantities).

Quantifying Composition: Chemical Units

The critical distinction between Mass and Volume.

Mole Fraction (x): Ratio of moles of a component to total moles. ($x_1 + x_2 + \dots = 1$)

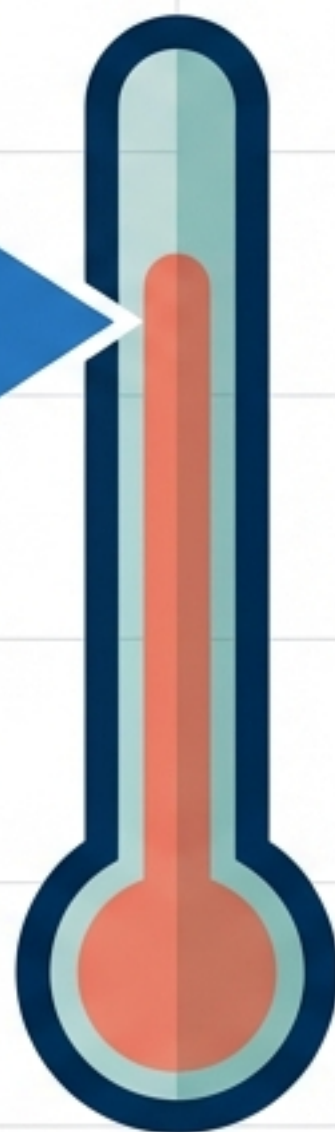
Molarity (M)

$$M = \frac{n}{V}$$

n = Moles of Solute
 V = Volume of Solution (Litres)



Temperature Dependent: Volume expands or contracts with heat, altering the Molarity.



Molality (m)

$$m = \frac{n}{\text{Mass(solvent)}}$$

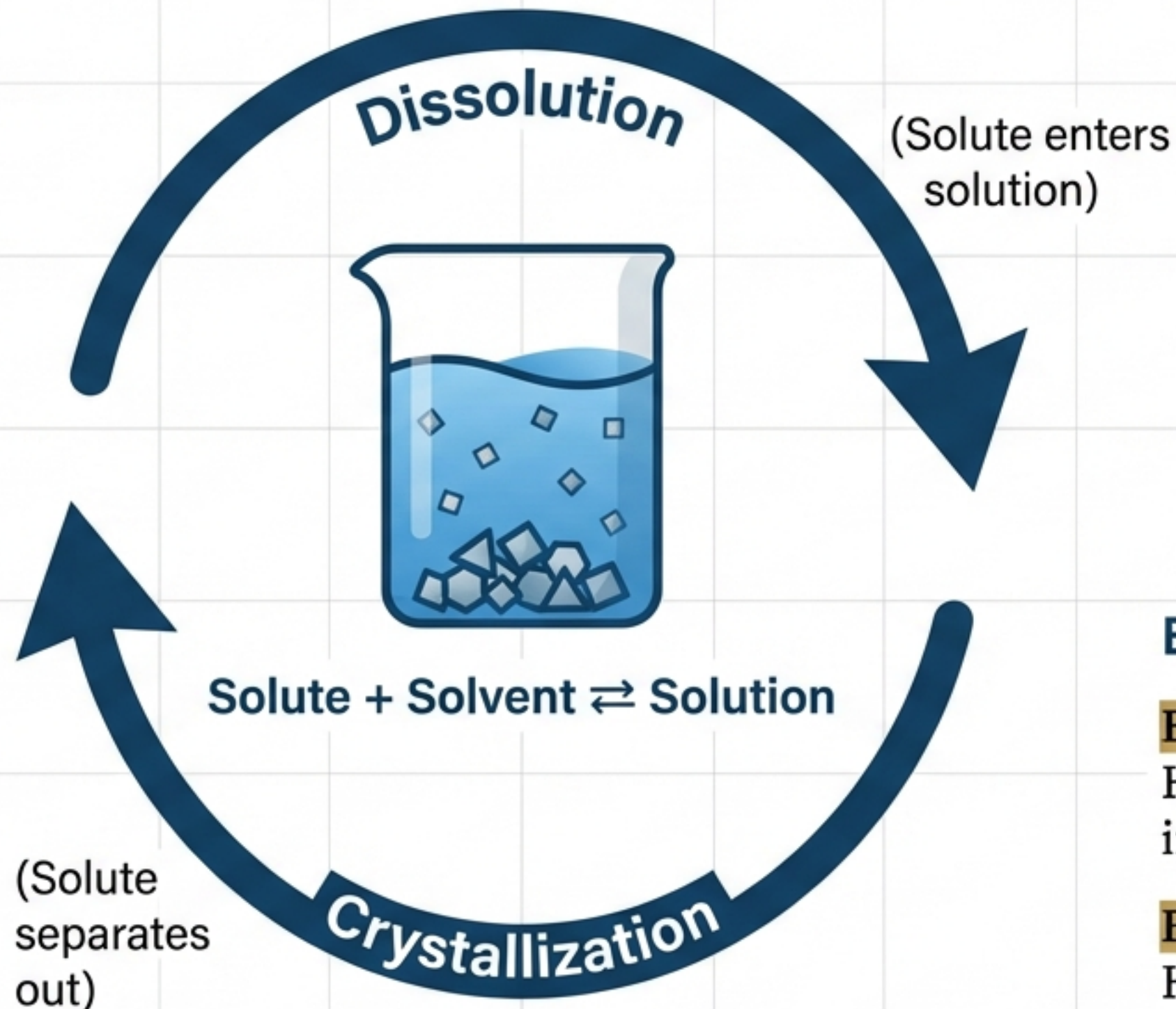
n = Moles of Solute
Mass = Mass of Solvent (kg)



Temperature Independent: Mass does not change with heat. Reliable for varying conditions.

Solubility & Dynamic Equilibrium

The rule of thumb: Like Dissolves Like.



Polarity Rules:

Polar solutes dissolve in polar solvents.

Non-polar solutes dissolve in non-polar solvents.

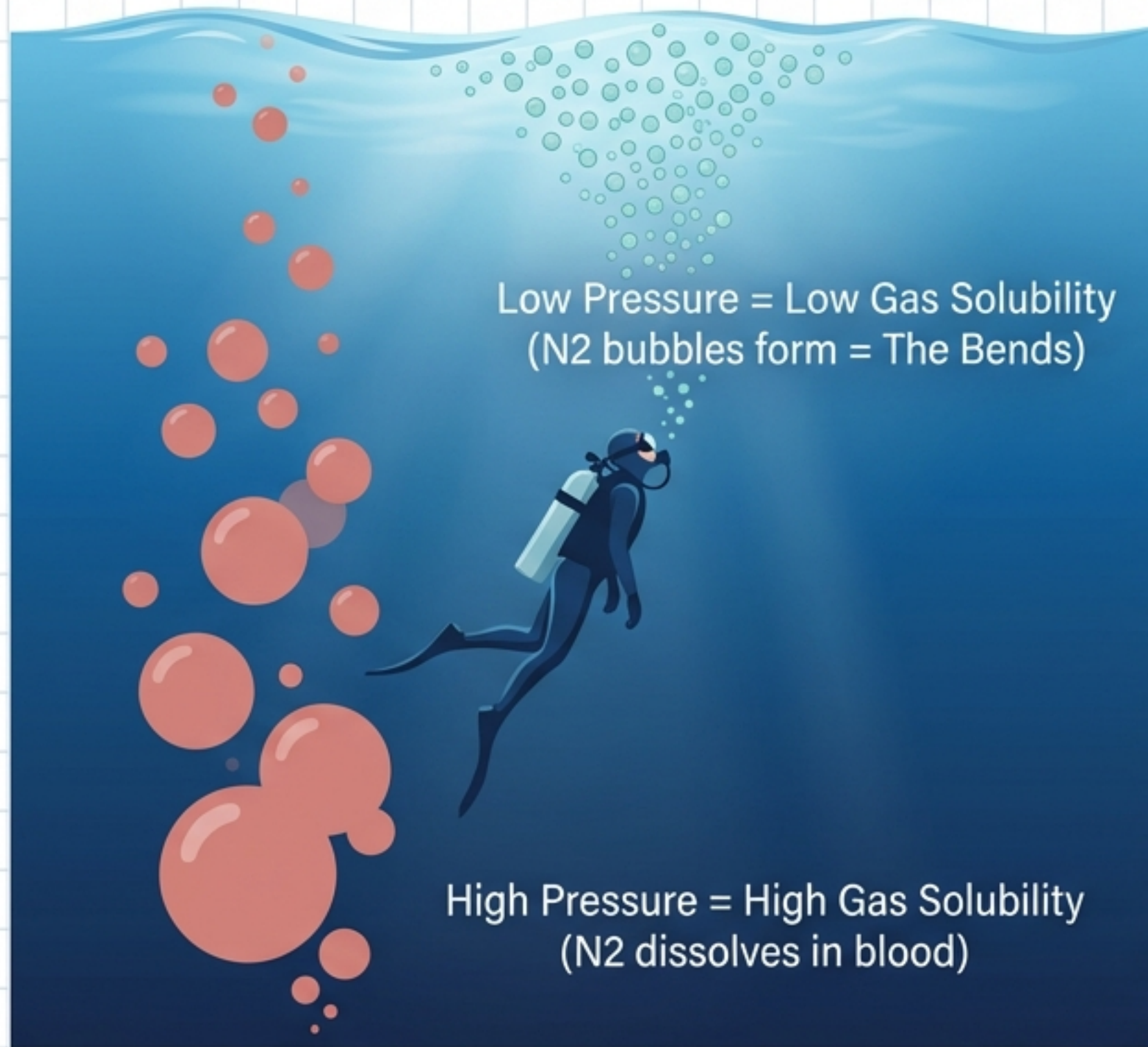
Example: Naphthalene dissolves in benzene, not water.

Effect of Temperature:

Endothermic ($\Delta_{\text{sol}} H > 0$): Heat is **absorbed**. Solubility increases as temp rises.

Exothermic ($\Delta_{\text{sol}} H < 0$): Heat is **released**. Solubility decreases as temp rises.

Gases in Liquids: Henry's Law



Scuba tanks dilute air with Helium to avoid this risk.

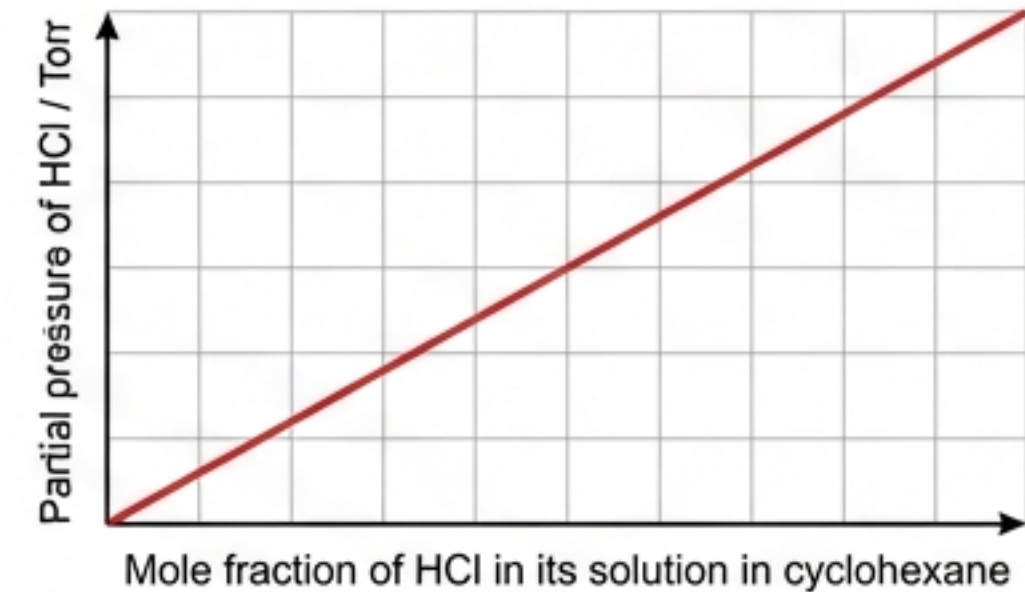
Henry's Law: The solubility of a gas is directly proportional to its partial pressure.

$$p = K_H \cdot x$$

p = partial pressure

x = mole fraction

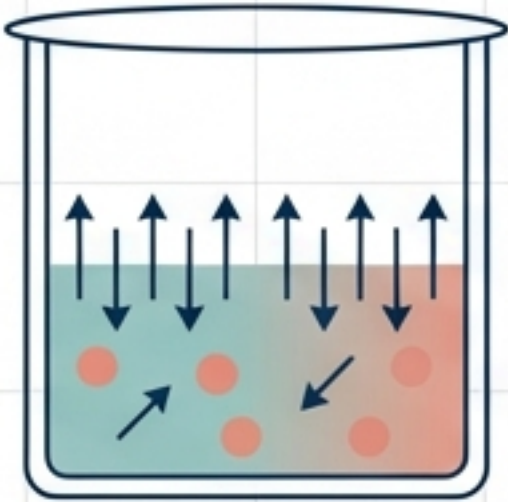
K_H = Henry's constant



! Key Insight: Higher K_H = Lower Solubility
Fish prefer cold water because K_H for Oxygen is lower at low temps (meaning Solubility is higher).

Vapour Pressure of Liquid Solutions

Raoult's Law for volatile components



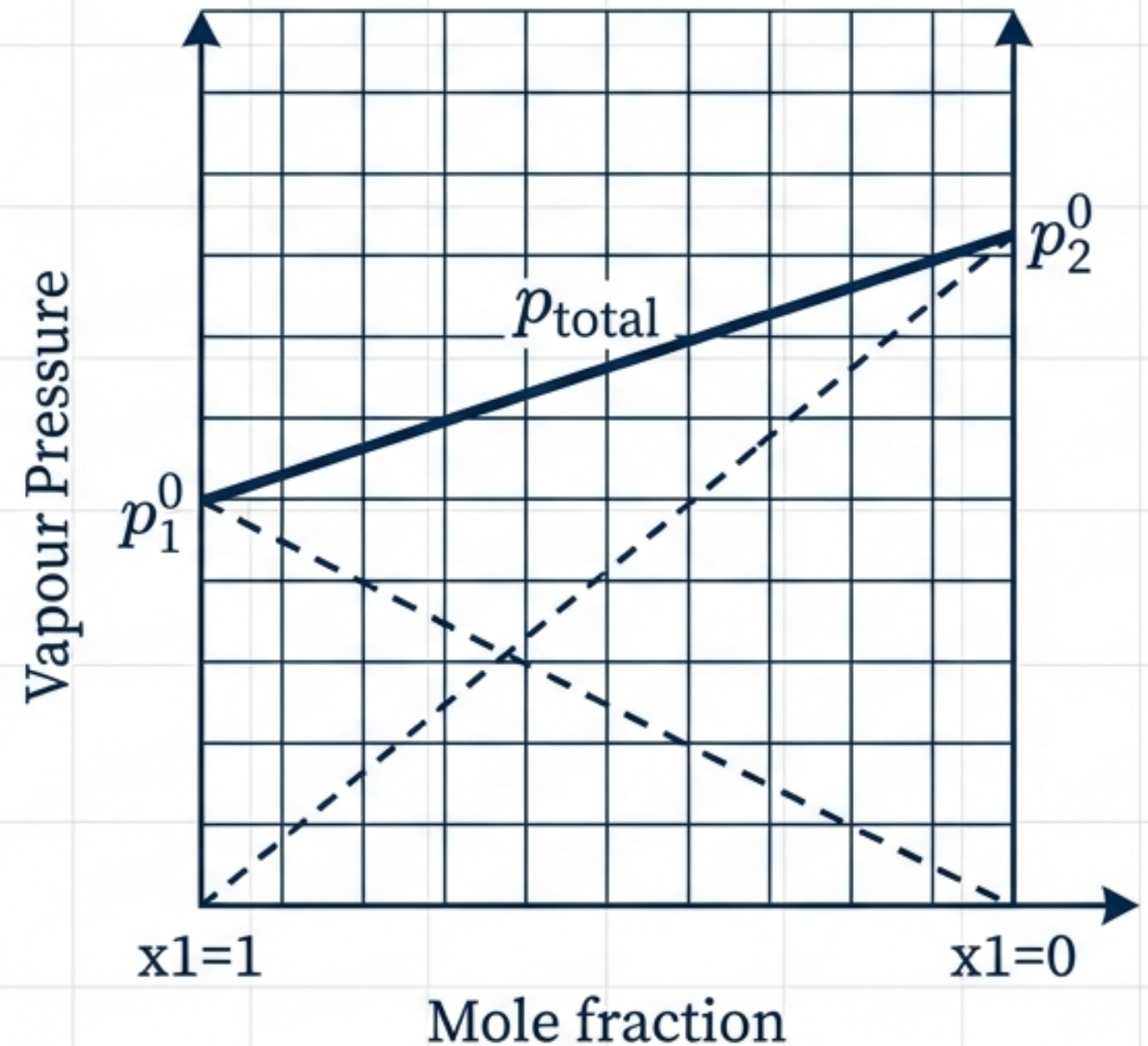
Dynamic Equilibrium:
Rate of evaporation = Rate of condensation

Raoult's Law: Partial pressure is proportional to mole fraction

$$p_1 = p_1^0 \cdot x_1$$

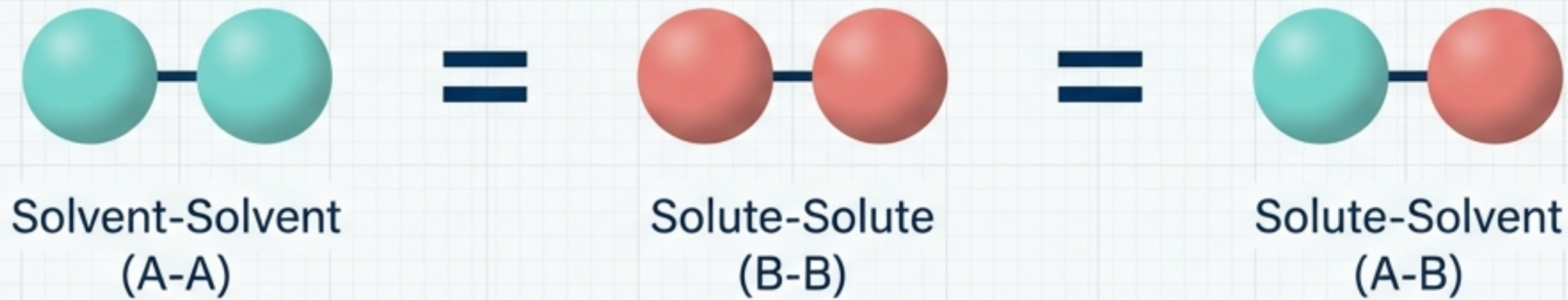
$$p_2 = p_2^0 \cdot x_2$$

$$p_{\text{total}} = p_1 + p_2$$



The Ideal Solution

When molecular interactions are perfectly balanced.



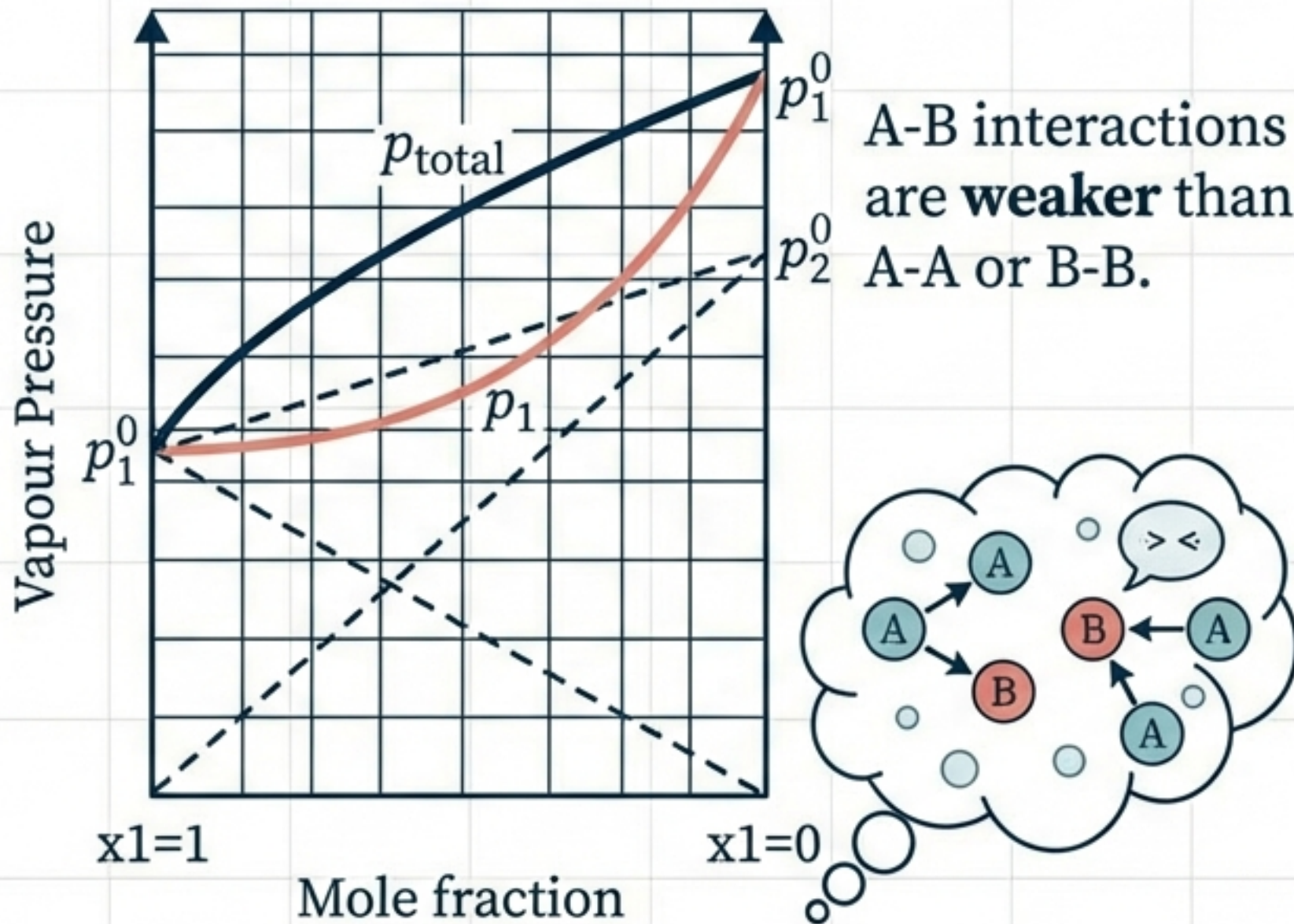
Interaction Strength: $A-A \approx B-B \approx A-B$

1. **Raoult's Law:** Obeyed over entire concentration range.
2. **Enthalpy of Mixing ($\Delta_{\text{mix}} H$) = 0:** No heat absorbed or released.
3. **Volume of Mixing ($\Delta_{\text{mix}} V$) = 0:** Total volume is the sum of components.

Examples: n-Hexane + n-Heptane | Benzene + Toluene | Bromoethane + Chloroethane

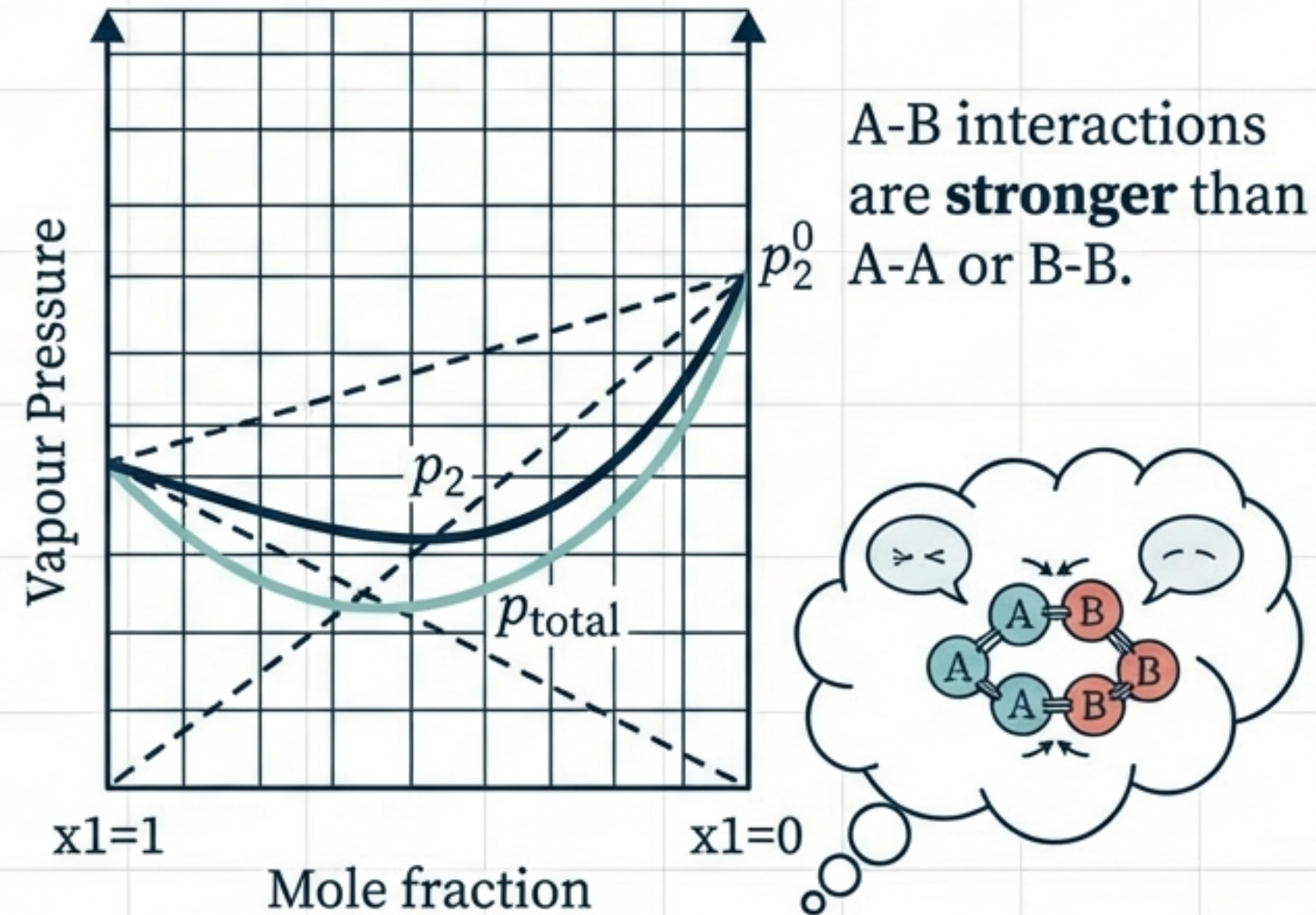
Non-Ideal Solutions: The Deviations

Positive Deviation



- Higher Vapour Pressure.
- Ethanol + Acetone.

Negative Deviation



- Lower Vapour Pressure.
- Phenol + Aniline.

Azeotropes: The Distillation Limit

Liquid mixtures that boil at a constant temperature like a pure substance.



Minimum Boiling Azeotrope

- Linked to: Large Positive Deviation.
- Boiling point is lower than either component.
- Example: 95% **Ethanol** / 5% **Water**.

Cannot separate ethanol beyond 95% by standard distillation.

Maximum Boiling Azeotrope

- Linked to: Large Negative Deviation.
- Boiling point is higher than either component.
- Example: 68% **Nitric Acid** / 32% **Water**
(Boils at 393.5 K).

Colligative Properties

The Democracy of Molecules



Properties that depend on the **number** of solute particles relative to the total number of particles, irrespective of their chemical nature.



Relative Lowering of Vapour Pressure

The presence of non-volatile solutes lowers the vapour pressure of the solvent.



Elevation of Boiling Point

The boiling point of a solution is higher than that of the pure solvent.



Depression of Freezing Point

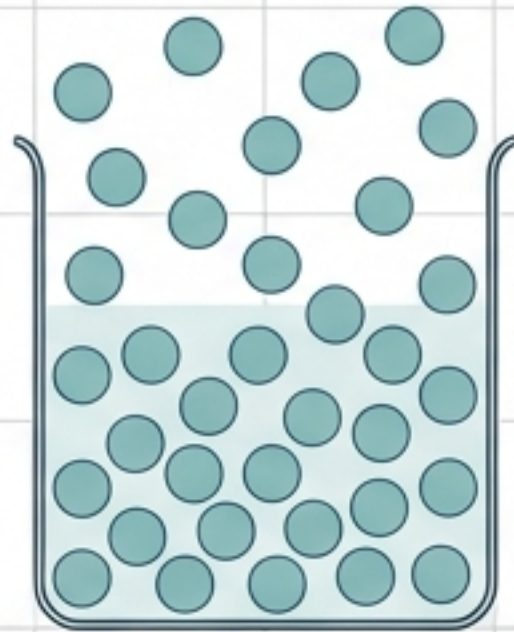
The freezing point of a solution is lower than that of the pure solvent.



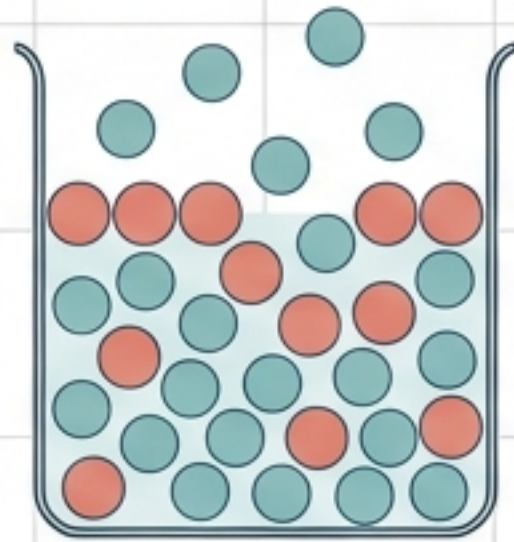
Osmotic Pressure

The pressure required to stop the flow of solvent across a semipermeable membrane.

1. Relative Lowering of Vapour Pressure



Pure Solvent
(All blue spheres)



Solution
(Mix of blue and non-volatile solute spheres)

The Mechanism:

Non-volatile solute particles physically occupy the surface area. Fewer solvent molecules can escape.

The Formula:

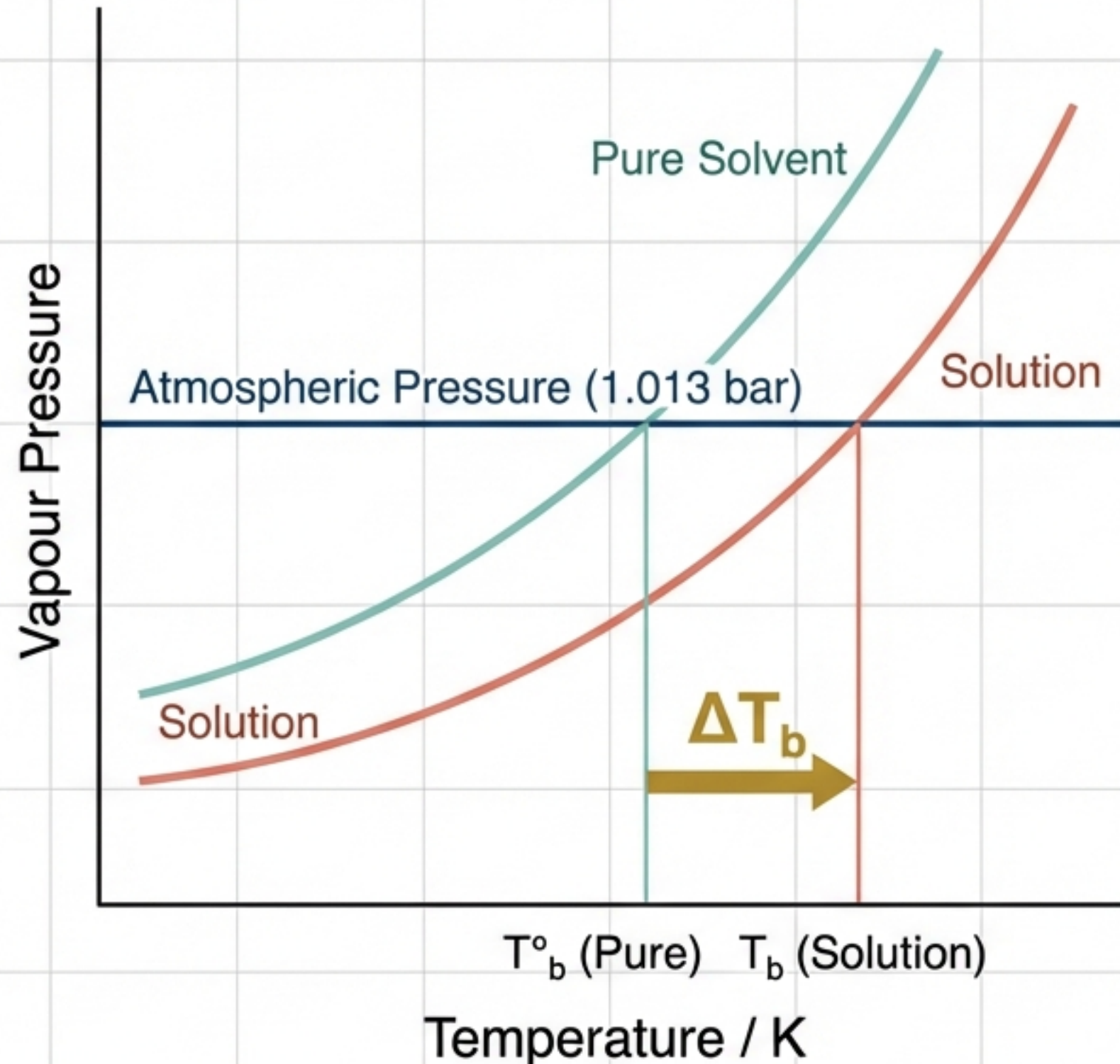
$$\frac{p_1^0 - p_1}{p_1^0} = x_2$$

Relative Lowering = Mole Fraction of Solute

Application:

Determining Molar Mass (M_2) of unknown solutes.

2. Elevation of Boiling Point (ΔT_b)



The Concept:

Because vapour pressure is lowered, the solution must be heated to a higher temperature to reach atmospheric pressure.

The Formula:

$$\Delta T_b = K_b \cdot m$$

- K_b : Ebullioscopic Constant
(Water = 0.52 K kg/mol)
- m : Molality

Example:

Dissolving 1 mole of solute in 1kg of water raises the Boiling Point to 373.67 K.

3. Depression of Freezing Point (ΔT_f)



Ethylene Glycol (Antifreeze)
allows cars to run at -17.6°C .

The Concept:

Solute particles disrupt the formation of the solvent's crystal lattice. Equilibrium between solid and liquid is reached at a lower temperature.

The Formula:

$$\Delta T_f = K_f \cdot m$$

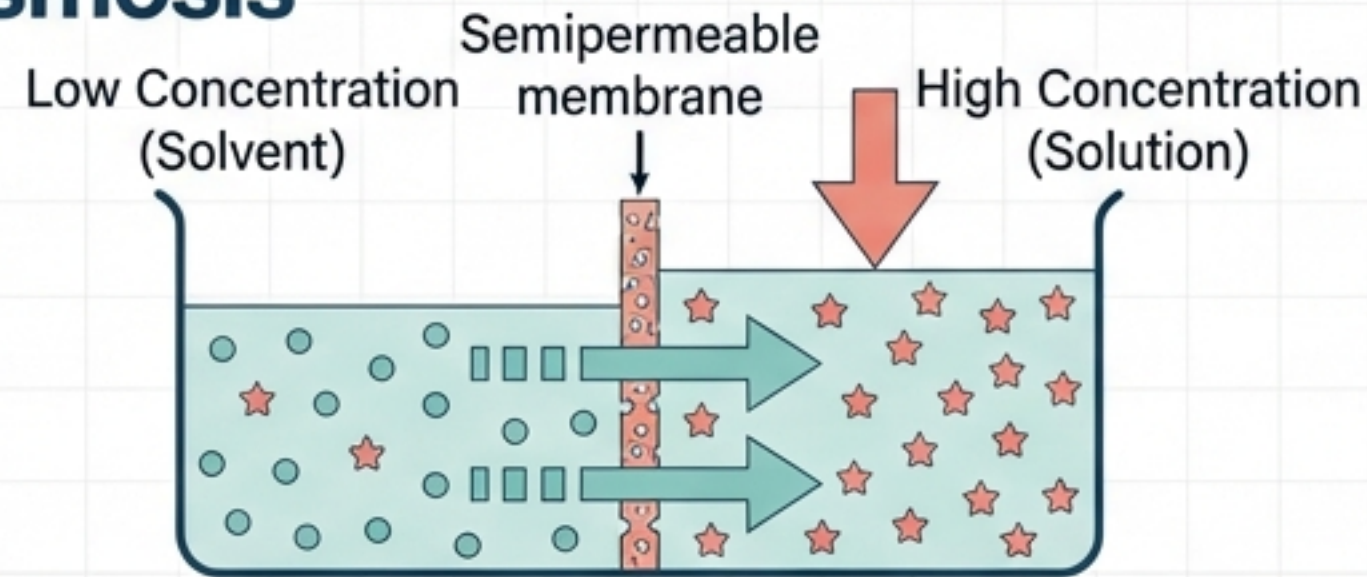
- K_f : Cryoscopic Constant
- m : Molality

Application:

De-icing roads with salt (NaCl) and car cooling systems.

4. Osmosis & Summary

Osmosis



Preserving pickles in brine
(salt withdraws water)



IV Fluids must be
isotonic with blood.

Osmotic Pressure (Π) is the pressure required to stop this flow.

Key Formula Cheat Sheet

Henry's Law

$$p = K_H \cdot x$$

p : Partial Pressure
 K_H : Henry's Law Constant
 x : Mole Fraction (gas)

Raoult's Law

$$p_1 = p_1^0 \cdot x_1$$

p_1 : Vapour Pressure (solution)
 p_1^0 : Vapour Pressure (pure)
 x_1 : Mole Fraction (solvent)

Boiling Point

$$\Delta T_b = K_b \cdot m$$

ΔT_b : Boiling Point Elevation
 K_b : Ebullioscopic Constant
 m : Molality

Freezing Point

$$\Delta T_f = K_f \cdot m$$

ΔT_f : Freezing Point Depression
 K_f : Cryoscopic Constant
 m : Molality

Understanding solutions allows us to control processes from biological survival to industrial manufacturing.